

PRODUCT REQUIREMENTS DOCUMENT (PRD)

1. PRODUCT OVERVIEW

Working Name

InviteIQ

Product Type

Mobile-first web application for secure invite-only event management with integrated guest verification, seating, commerce, and logistics coordination.

Core Objective

Replace fragmented event execution tools (WhatsApp, Excel, paper lists) with a unified system for:

- Controlled guest access
 - Structured RSVP handling
 - Seating automation
 - Real-time check-in
 - Event-linked commerce (Aso Ebi)
 - Delivery coordination for purchased items
-

2. MVP SCOPE

This MVP includes event execution, guest control, seating intelligence, offline check-in, and lightweight commerce.

2.1 CORE EVENT SYSTEM

Included Features

- Event creation (name, venue, time, capacity)
 - Guest list upload (CSV, manual entry, bulk paste)
 - Guest categorization (VIP, regular, family, etc.)
 - Invite generation (unique link + QR code per guest)
 - RSVP management (Yes / No / Maybe / Waitlist)
 - Event dashboard for organizers and planners
 - Role-based access control (Organizer, Planner, Security, Guest, Admin)
-

2.2 GUEST INVITE AND RSVP SYSTEM

Guest Flow

- Guest receives unique invitation link
- Views event details on mobile-first page
- Submits RSVP response
- Receives QR-based digital pass upon acceptance
- Optional view of assigned seat/table

System Rules

- Each guest is uniquely identified by secure token
 - RSVP updates reflected in real-time dashboard
 - Changes in RSVP may trigger seating reallocation
-

2.3 SEATING SYSTEM

Manual Seating (Baseline)

- Drag-and-drop table layout editor
 - Table creation with capacity settings
 - Manual assignment of guests to seats
 - Real-time visualization of seating chart
-

AI SEATING OPTIMIZATION (MVP FEATURE)

Purpose

Automatically generate structured seating arrangements based on constraints.

Inputs

- Guest categories (VIP, family, friends, corporate, etc.)
- Table capacities
- RSVP confirmations
- Optional constraints (grouping, separation rules)

Output

- Suggested seating arrangement
- Table assignments per guest
- Conflict warnings
- Manual override enabled

Logic Approach (Non-ML MVP)

Deterministic rule-based system:

1. Group guests by category
2. Prioritize VIP placement rules
3. Apply grouping constraints
4. Fill tables using balanced distribution
5. Resolve conflicts iteratively

Constraint

No machine learning required in MVP. Must remain explainable and deterministic.

2.4 CHECK-IN SYSTEM

Core Functionality

- QR code scanning for guest verification
 - Real-time check-in marking
 - Duplicate scan prevention
 - Attendance tracking per event
-

OFFLINE-FIRST QR CHECK-IN (MVP FEATURE)

Objective

Ensure check-in works in venues with poor or no internet connectivity.

Design Model

Pre-Event Sync

- Full guest list is cached locally on security device
- Includes QR tokens and validation dataset

Offline Operation

- QR scanning validates against local dataset
- Check-ins stored locally in a queue

Sync Process

- When internet is restored:
 - queued check-ins are pushed to Supabase
 - server validates duplicates
 - conflict logs are generated if needed

Conflict Rules

- Server is authoritative source
 - Duplicate check-ins are rejected server-side
 - Local data is periodically refreshed per session
-

2.5 MESSAGING SYSTEM

Features

- Bulk messaging (email/SMS/WhatsApp abstraction)
 - Segmentation filters:
 - RSVP status
 - Guest category (VIP, regular)
 - Scheduled reminders and follow-ups
 - Thank-you message automation (basic template system)
-

2.6 ASO EBI COMMERCE SYSTEM

Objective

Enable event organizers to sell coordinated attire directly within event workflow.

PRODUCT MODULE

Features

- Aso Ebi product creation per event:
 - Name
 - Fabric type
 - Price
 - Available sizes
 - Order deadline
 - Guest-facing purchase interface inside invite page
-

ORDER FLOW

1. Guest selects Aso Ebi item
2. Chooses size and quantity

3. Places order linked to event and guest profile
 4. Order stored with payment status
-

PAYMENT SYSTEM

- Integrated payment gateway (Paystack recommended)
 - Single-payment flow (no split settlement in MVP)
 - Organizer receives aggregated order summary
-

2.7 LOGISTICS COORDINATION SYSTEM

Purpose

Handle delivery of Aso Ebi items to guests.

FEATURES

Delivery Details Collection

- Guest provides delivery address
- Phone number confirmation
- Delivery preference notes

Logistics Assignment

- Organizer assigns logistics partner manually per event
- Orders routed into partner queue

Delivery Tracking Status

- Pending pickup
 - In transit
 - Delivered
 - Failed delivery
-

IMPLEMENTATION MODEL

- No external logistics API required in MVP
 - Status updates handled manually via admin dashboard or CSV import
-

2.8 ANALYTICS SYSTEM

Metrics Included

- Total invited guests
- RSVP breakdown
- Attendance rate
- No-show tracking
- Check-in timestamps
- Aso Ebi order volume per event

Implementation

- Supabase SQL aggregation queries
 - Optional materialized views for performance
-

2.9 USER ROLES

Organizer

- Full event creation and configuration
- Guest management
- Seating control
- Messaging
- Commerce setup

Planner

- Guest list management
- Seating assignments
- Messaging execution
- Event monitoring

Security (Door Staff)

- QR scanning interface
- Check-in validation
- Offline sync support

Guest

- RSVP handling
- View invite and digital pass
- Purchase Aso Ebi

- View seating (if enabled)

Admin

- System oversight
 - User and event moderation
 - Platform configuration
-

3. SYSTEM ARCHITECTURE

Frontend

- Next.js (mobile-first web app)
 - TailwindCSS
 - React Query for data sync
 - QR scanning via browser camera APIs
-

Backend (Supabase)

- PostgreSQL database
 - Authentication (email + magic link optional)
 - Row Level Security (RLS)
 - Realtime subscriptions
 - Storage (CSV uploads, assets)
-

External Services

- Paystack (payments)
 - Resend / SendGrid (email)
 - Twilio / Termii (SMS/WhatsApp)
-

4. DATABASE STRUCTURE (CORE TABLES)

Events

- event metadata and configuration

Guests

- guest identity, RSVP, QR token

Tables & Seats

- seating layout and assignments

Check-ins

- verified attendance records

Messages

- communication logs

Aso Ebi Products

- event-linked merchandise

Orders

- guest purchases

Delivery Orders

- logistics tracking

Audit Logs

- system activity tracking
-

5. SECURITY MODEL

QR Security

- Signed unique tokens per guest
 - One-time check-in enforcement
 - Server-side validation for all scans
-

RLS POLICY DESIGN

- Guests restricted to own data only
 - Security role limited to check-in actions
 - Planner access scoped per event
 - Organizer owns event-level permissions
-

6. PERFORMANCE REQUIREMENTS

- QR scan validation: < 500ms
- RSVP update latency: < 1 second
- Real-time dashboard sync: < 2 seconds
- Support event sizes up to 10,000 guests (MVP ceiling)

7. EDGE CASE HANDLING

- Duplicate QR scans → rejected with audit log
 - Invalid QR → flagged immediately
 - Walk-in guests → manual guest creation flow
 - RSVP change after seating → triggers seat reallocation
 - Offline check-in sync conflicts → server override
 - Payment failure → order remains pending
-

8. KEY USER FLOWS

Organizer Flow

Create event → upload guests → configure seating → send invites → monitor RSVP → manage check-in → review analytics → handle Aso Ebi orders

Guest Flow

Receive invite → RSVP → receive QR pass → optionally purchase Aso Ebi → attend event → get checked in

Security Flow

Login → load cached guest list → scan QR → validate locally → sync check-ins when online

9. SUCCESS METRICS

- 90%+ RSVP completion rate per event
- Check-in processing < 1 second per guest
- Zero dependency on paper-based guest lists
- Successful offline check-in operation in low-connectivity environments
- At least 1% conversion to Aso Ebi purchases per eligible event